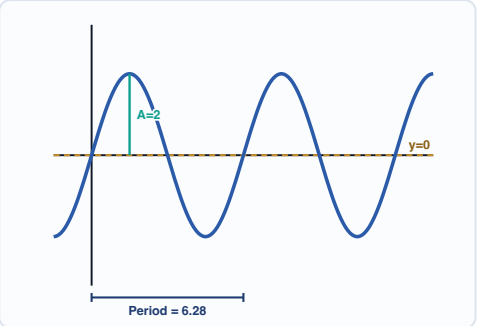
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Trigonometry, Identities & Polar Coordinates Practice Exercises

1 MCQ

A point has polar coordinates $(r, \theta) = (6, \frac{2\pi}{3})$. What are the Cartesian coordinates (x, y) of this point?



A $(-3, 3\sqrt{3})$

B $(-3\sqrt{3}, 3)$

C $(3, -3\sqrt{3})$

D $(3\sqrt{3}, -3)$

2 MCQ

In $\triangle ABC$, the length of side AB is x, the length of side BC is $x + 3$, and the length of side AC is 7. If $\angle B = 60^\circ$, what is the value of x?

A 5

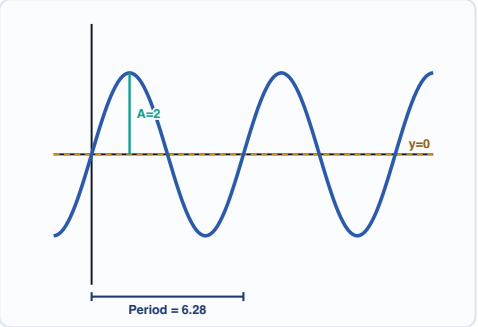
B 8

C 4

D 6

3 MCQ

In the polar coordinate plane, points A and B lie on the curve $r = 4\sin(2\theta)$ such that their polar angles satisfy $0 < \theta_A < \theta_B < \frac{\pi}{2}$ and $\theta_B - \theta_A = \frac{\pi}{3}$. What is the maximum possible area of $\triangle OAB$, where O is the origin?



A $\sqrt{3}$

B 2

C $2\sqrt{3}$

D $4\sqrt{3}$

4 MCQ

A point in the polar coordinate system has coordinates $(r, \theta) = (6, \frac{5\pi}{6})$. Which of the following represents the Cartesian coordinates (x, y) of this point?

A $(-3\sqrt{3}, 3)$

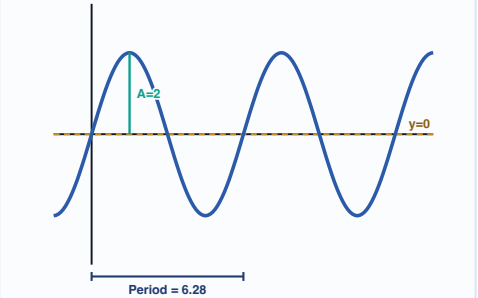
B $(-3, 3\sqrt{3})$

C $(3\sqrt{3}, -3)$

D $(-3\sqrt{3}, -3)$

5 MCQ

In $\triangle ABC$, the lengths of sides AB and BC are 5 and 8, respectively, and the measure of $\angle B$ is 60°. What is the length of the altitude from vertex B to side AC?



A $\frac{20\sqrt{3}}{7}$

B $\frac{40\sqrt{3}}{7}$

C $\frac{10\sqrt{3}}{7}$

D $\frac{20}{7}$

6 MCQ

In the polar coordinate system, point A has polar coordinates (r_A, θ_A) lying on the curve $r = 6\sin\theta$, and point B has polar coordinates (r_B, θ_B) lying on the curve $r = 8\cos\theta$, with $r_A > 0$ and $r_B > 0$. If $\theta_A - \theta_B = \frac{\pi}{6}$, what is the maximum possible area of $\triangle OAB$, where O is the pole (origin)?

A 6

B $6\sqrt{3}$

C 9

D 12

7 MCQ

A point in the polar coordinate plane has coordinates $(r, \theta) = (4, \frac{5\pi}{6})$. Which of the following gives the rectangular coordinates (x, y) of this point?

A $(-2\sqrt{3}, 2)$

B $(-2, 2\sqrt{3})$

C $(2, -2\sqrt{3})$

D $(2\sqrt{3}, -2)$

8 MCQ

In $\triangle ABC$, the side lengths are $AB = 5$ and $BC = 4$, and $\angle B$ is an acute angle. If $\cos(2B) = -\frac{31}{32}$, what is the length of side AC?

A 6

B $\sqrt{46}$

C $\sqrt{41}$

D $\sqrt{31}$

9 MCQ

In the polar coordinate plane, a curve is defined by the equation $r = 6\cos\theta + 8\sin\theta$. Two rays emanating from the pole at angles $\theta = \alpha$ and $\theta = \alpha + \frac{\pi}{2}$ intersect the curve at points $P \neq O$ and $Q \neq O$, respectively, where O denotes the pole. Assuming α is chosen such that both P and Q have positive radial coordinates, what is the maximum possible area of $\triangle OPQ$?

A 25

B 24

C 48

D 50

10 MCQ

In the standard (x, y)-coordinate plane, what are the rectangular coordinates of the point with polar coordinates $(r, \theta) = (6, \frac{5\pi}{6})$?

A $(-3, 3\sqrt{3})$

B $(3\sqrt{3}, -3)$

C $(-3\sqrt{3}, 3)$

D $(-3\sqrt{3}, -3)$

11 MCQ

Find $\sin(30^\circ)$.

A 1

B $-\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{1}{2}$

12 MCQ

Find $\cos(180^\circ)$.

A 0

B -1

C $-\frac{\sqrt{2}}{2}$

D $\frac{\sqrt{2}}{2}$

13 MCQ

Calculate $\cos(45^\circ)$.

A $\frac{\sqrt{2}}{2}$

B $\frac{1}{2}$

C 1

D $\frac{\sqrt{3}}{2}$

14 MCQ

Calculate $\sin(270^\circ)$.

A -1

B $\frac{\sqrt{3}}{2}$

C 0

D $\frac{1}{2}$

15 MCQ

Compute $\cos(300^\circ)$.

A 0

B $\frac{\sqrt{2}}{2}$

C $-\frac{1}{2}$

D $\frac{1}{2}$

16 MCQ

Compute $\sin(225^\circ)$.

A 0

B $-\frac{\sqrt{2}}{2}$

C $-\frac{1}{2}$

D $-\frac{\sqrt{3}}{2}$

17 MCQ

Deduce $\sin(210^\circ)$.

A 1

B $-\frac{1}{2}$

C $\frac{\sqrt{2}}{2}$

D -1

18 MCQ

Obtain $\sin(180^\circ)$.

A $\frac{1}{2}$

B $-\frac{\sqrt{2}}{2}$

C $\frac{\sqrt{2}}{2}$

D 0

19 MCQ

Deduce $\cos(90^\circ)$.

A $-\frac{1}{2}$

B 0

C -1

D 1

20 MCQ

Obtain $\cos(120^\circ)$.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C -1

D $-\frac{\sqrt{2}}{2}$

21 MCQ

Identify $\cos(30^\circ)$.

A $-\frac{\sqrt{2}}{2}$

B $\frac{\sqrt{3}}{2}$

C $-\frac{1}{2}$

D $-\frac{\sqrt{3}}{2}$

22 MCQ

Identify $\sin(90^\circ)$.

A $-\frac{\sqrt{3}}{2}$

B 1

C $-\frac{1}{2}$

D -1

23 MCQ

Establish $\sin(0^\circ)$.

A $\frac{\sqrt{3}}{2}$

B 0

C -1

D $-\frac{\sqrt{3}}{2}$

24 MCQ

Determine $\sin(300^\circ)$.

A $\frac{\sqrt{3}}{2}$

B $-\frac{\sqrt{3}}{2}$

C $-\frac{\sqrt{2}}{2}$

D 1

25 MCQ

Establish $\cos(240^\circ)$.

A $-\frac{1}{2}$

B 1

C -1

D $\frac{1}{2}$